

An Information-Theoretic Perspective on Covert Uplink Communication in FD-ISAC Systems

System Model. We consider a simplified single-antenna full-duplex (FD) system that isolates the fundamental interaction between an uplink covert transmission and a concurrent downlink waveform. Alice communicates covertly with an FD base station (BS), while a passive warden Willie attempts to determine whether Alice is active. During a block of n channel uses, the BS simultaneously transmits an independent Gaussian downlink waveform

$$S_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, P_d), \quad i = 1, \dots, n, \quad (1)$$

where P_d is fixed. All channel coefficients are assumed constant over the block and known. Willie knows the distribution and power of S^n , but not its realization.

After self-interference cancellation, the BS observes

$$Y_i = h_{ab}X_i + \tilde{N}_{b,i}, \quad \tilde{N}_{b,i} \sim \mathcal{CN}(0, N_b(P_d)), \quad (2)$$

where $N_b(P_d) \triangleq \sigma_b^2 + \beta|h_{\text{SI}}|^2 P_d$ models receiver noise and residual self-interference. Here h_{ab} is the Alice–BS channel, h_{SI} is the self-interference channel, and $\beta \geq 0$ denotes the residual self-interference factor.

Willie’s observation is

$$Z_i = g_{aw}X_i + g_{bw}S_i + N_{w,i}, \quad N_{w,i} \sim \mathcal{CN}(0, \sigma_w^2), \quad (3)$$

where g_{aw} and g_{bw} denote the Alice–Willie and BS–Willie channels, respectively. Defining

$$V_i \triangleq g_{bw}S_i + N_{w,i} \sim \mathcal{CN}(0, N_w(P_d)), \quad N_w(P_d) \triangleq \sigma_w^2 + |g_{bw}|^2 P_d, \quad (4)$$

the surveillance channel becomes $Z_i = g_{aw}X_i + V_i$.

Code Definition. For blocklength n , Alice employs a fixed (n, M_n) codebook

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\} \subset \mathbb{C}^n, \quad (5)$$

with a uniformly distributed message $M \sim \text{Unif}\{1, \dots, M_n\}$. The encoder maps $M = m$ to the deterministic codeword $x^n(m)$, subject to the per-codeword power constraint

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad \forall m. \quad (6)$$

The BS decoder $\phi_n : \mathbb{C}^n \rightarrow \{1, \dots, M_n\}$ produces $\hat{M} = \phi_n(Y^n)$, and reliability requires

$$P_e^{(n)} \triangleq \Pr(\hat{M} \neq M) \rightarrow 0. \quad (7)$$

Willie’s Hypothesis Test and Coverttness. Under Alice’s silence,

$$\mathcal{H}_0 : \quad Z^n = V^n, \quad P_0^{(n)} = \mathcal{CN}(\mathbf{0}, N_w(P_d)\mathbf{I}_n). \quad (8)$$

Under $\mathcal{H}_1$, conditioned on message $M = m$ and the fixed codebook,

$$Z^n = g_{aw}x^n(m) + V^n, \quad (9)$$

and therefore

$$P_{Z^n|M=m, \mathcal{H}_1, \mathcal{C}_n} = \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (10)$$

Since Willie does not know M , the actual distribution induced by the fixed codebook is the Gaussian mixture

$$P_{1, \mathcal{C}_n}^{(n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \mathcal{CN}(g_{aw}x^n(m), N_w(P_d)\mathbf{I}_n). \quad (11)$$

We impose the coverttness constraint

$$D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta, \quad (12)$$

for a prescribed $\delta > 0$. The fundamental objective is to characterize the maximum codebook size

$$M_n^*(\delta) = \max_{\mathcal{C}_n, \phi_n} \left\{ M_n : P_e^{(n)} \rightarrow 0, D(P_{1, \mathcal{C}_n}^{(n)} \| P_0^{(n)}) \leq \delta \right\}, \quad (13)$$

and, in particular, the asymptotic growth of $\log M_n^*(\delta)$ as $n \rightarrow \infty$ and its dependence on the concurrent downlink power P_d .

Theorem 1 (Achievable Square-Root Law for FD-ISAC). *Assume $g_{aw} \neq 0$, $P_a > 0$ is fixed, and*

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}. \quad (14)$$

For every prescribed $\delta > 0$, there exists a sequence of deterministic fixed codebooks satisfying the per-codeword power constraint, vanishing decoding error probability, and

$$D\left(P_{1,\mathcal{C}_n}^{(n)} \parallel P_0^{(n)}\right) \leq \delta. \quad (15)$$

Moreover, the maximum admissible codebook size satisfies

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \quad (16)$$

Thus the achievable covert information grows on the square-root scale, $\log M_n = \Theta(\sqrt{n})$.

Physical Interpretation. The achievable square-root coefficient depends on the concurrent downlink power P_d through

$$\frac{N_w(P_d)}{N_b(P_d)} = \frac{\sigma_w^2 + |g_{bw}|^2 P_d}{\sigma_b^2 + \beta |h_{SI}|^2 P_d}. \quad (17)$$

Increasing P_d therefore has two competing effects: it increases the uncertainty at Willie through the BS–Willie link, while simultaneously increasing the residual self-interference at the BS. In particular, the achievable coefficient increases with P_d if

$$|g_{bw}|^2 \sigma_b^2 > \beta |h_{SI}|^2 \sigma_w^2, \quad (18)$$

decreases with P_d when the inequality is reversed, and is independent of P_d at the square-root order when equality holds. The downlink power also affects the feasibility condition in Theorem 1 through the relative effective qualities of the Alice–BS and Alice–Willie channels.

Proof. Random-Coding Construction. For each blocklength n , let $Q_n > 0$ and $M_n \in \mathbb{N}$. Generate the random codebook

$$\mathcal{C}_n = \{X^n(1), \dots, X^n(M_n)\}, \quad X_i(m) \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, Q_n), \quad (19)$$

independently over $i = 1, \dots, n$ and $m = 1, \dots, M_n$. A realization of $\mathcal{C}_n$ is denoted by

$$\mathcal{C}_n = \{x^n(1), \dots, x^n(M_n)\}.$$

For $M \sim \text{Unif}\{1, \dots, M_n\}$, Alice transmits $x^n(M)$. The BS employs the ML decoder

$$\phi_n(y^n) = \arg \min_{1 \leq m \leq M_n} \|y^n - h_{ab} x^n(m)\|^2. \quad (20)$$

To characterize the output distribution induced by the Gaussian codeword ensemble at Willie, recall that

$$Z_i = g_{aw} X_i + V_i, \quad V_i \sim \mathcal{CN}(0, N_w(P_d)).$$

Since X_i and V_i are independent Gaussian random variables,

$$Z_i \sim \mathcal{CN}(0, N_w(P_d) + |g_{aw}|^2 Q_n). \quad (21)$$

Independence across channel uses therefore gives the Gaussian reference distribution

$$\bar{P}_{1,n}^{(n)} \triangleq \mathcal{CN}(\mathbf{0}, [N_w(P_d) + |g_{aw}|^2 Q_n] \mathbf{I}_n). \quad (22)$$

Reference Coverttness and Power Scaling. We first evaluate the divergence between the Gaussian reference distribution in (17) and the innocent distribution $P_0^{(n)}$ in (8). For zero-mean proper complex Gaussian distributions,

$$D(\mathcal{CN}(\mathbf{0}, \Sigma_1) \parallel \mathcal{CN}(\mathbf{0}, \Sigma_0)) = \log \frac{\det \Sigma_0}{\det \Sigma_1} + \text{tr}(\Sigma_0^{-1} \Sigma_1) - n.$$

Hence,

$$D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}) = n \left[\frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \right]. \quad (23)$$

Imposing

$$D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right) \leq \delta \quad (24)$$

gives

$$0 \leq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} - \log\left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)}\right) \leq \frac{\delta}{n}. \quad (25)$$

Since $x - \log(1+x)$ is strictly increasing for $x \geq 0$ and vanishes only at $x = 0$, it follows that

$$Q_n \longrightarrow 0. \quad (26)$$

The expansion

$$x - \log(1+x) = \frac{x^2}{2} + O(x^3), \quad x \rightarrow 0,$$

therefore yields

$$D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right) = \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)). \quad (27)$$

Consequently,

$$Q_n \leq \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}} (1 + o(1)), \quad (28)$$

and hence

$$Q_n = O\left(n^{-1/2}\right). \quad (29)$$

Power-Sequence Selection. Fix an arbitrary $0 < \varepsilon < 1$. Motivated by (23), choose

$$Q_n = (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{\frac{2\delta}{n}}. \quad (30)$$

Substituting this choice into (22) gives

$$\begin{aligned} D\left(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}\right) &= \frac{n|g_{aw}|^4 Q_n^2}{2N_w^2(P_d)} (1 + o(1)) \\ &= (1 - \varepsilon)^2 \delta (1 + o(1)) < \delta \end{aligned} \quad (31)$$

for all sufficiently large n . Moreover,

$$\sqrt{n} Q_n \longrightarrow (1 - \varepsilon) \frac{N_w(P_d)}{|g_{aw}|^2} \sqrt{2\delta}, \quad (32)$$

and hence

$$Q_n = \Theta\left(n^{-1/2}\right). \quad (33)$$

Reliability at the BS. Under the Gaussian input $X_i \sim \mathcal{CN}(0, Q_n)$, Bob's channel in (2) induces

$$Y_i \sim \mathcal{CN}(0, N_b(P_d) + |h_{ab}|^2 Q_n).$$

Let $p_{Y|X}(\cdot|x)$ denote the conditional Gaussian density in (2), and let $p_{Y,n}$ denote the above output density. Define the single-letter information density by

$$i_{b,n}(x; y) \triangleq \log \frac{p_{Y|X}(y|x)}{p_{Y,n}(y)}, \quad (34)$$

and, by memorylessness,

$$i_{b,n}^{(n)}(X^n; Y^n) = \sum_{i=1}^n i_{b,n}(X_i; Y_i). \quad (35)$$

Its mean is the Gaussian-input mutual information

$$\begin{aligned} I_{b,n} &\triangleq \mathbb{E}[i_{b,n}(X_i; Y_i)] \\ &= \log\left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)}\right), \end{aligned} \quad (36)$$

while direct evaluation of the Gaussian information density gives

$$\mathbb{E}\left[i_{b,n}^{(n)}(X^n; Y^n)\right] = nI_{b,n}, \quad \text{Var}\left[i_{b,n}^{(n)}(X^n; Y^n)\right] = \frac{2n|h_{ab}|^2 Q_n}{N_b(P_d) + |h_{ab}|^2 Q_n}. \quad (37)$$

Since the ML decoder minimizes the average decoding error for a fixed codebook, by Feinstein's random-coding bound, for any $\tau_n > 0$,

$$\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}] \leq \Pr [\iota_{b,n}^{(n)}(X^n; Y^n) \leq \log M_n + \tau_n] + e^{-\tau_n}. \quad (38)$$

Define the reliability margin

$$\Delta_{b,n} \triangleq nI_{b,n} - \log M_n. \quad (39)$$

For $\Delta_{b,n} > 0$, choosing $\tau_n = \Delta_{b,n}/2$ and applying Chebyshev's inequality gives

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}] &\leq \frac{4 \text{Var}[\iota_{b,n}^{(n)}]}{\Delta_{b,n}^2} + e^{-\Delta_{b,n}/2} \\ &= \frac{8n|h_{ab}|^2 Q_n}{[N_b(P_d) + |h_{ab}|^2 Q_n] \Delta_{b,n}^2} + e^{-\Delta_{b,n}/2}. \end{aligned} \quad (40)$$

Since $Q_n = \Theta(n^{-1/2})$, the variance term is $\Theta(\sqrt{n})$. Hence

$$\frac{\Delta_{b,n}}{n^{1/4}} \longrightarrow \infty \quad (41)$$

is sufficient to ensure

$$\mathbb{E}_{\mathbf{C}_n} [P_e^{(n)}] \longrightarrow 0. \quad (42)$$

Moreover,

$$\begin{aligned} nI_{b,n} &= n \log \left(1 + \frac{|h_{ab}|^2 Q_n}{N_b(P_d)} \right) \\ &= \frac{|h_{ab}|^2}{N_b(P_d)} nQ_n + O(nQ_n^2) \\ &= (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + O(1). \end{aligned} \quad (43)$$

Resolvability at Willie. Let $p_{Z|X}(z|x)$ denote the conditional density of Willie's channel,

$$p_{Z|X}(z|x) = \frac{1}{\pi N_w(P_d)} \exp \left(-\frac{|z - g_{aw}x|^2}{N_w(P_d)} \right), \quad (44)$$

and let $\bar{p}_{1,n}(z)$ denote the single-letter density associated with the Gaussian reference distribution in (17),

$$\bar{p}_{1,n}(z) = \frac{1}{\pi[N_w(P_d) + |g_{aw}|^2 Q_n]} \exp \left(-\frac{|z|^2}{N_w(P_d) + |g_{aw}|^2 Q_n} \right). \quad (45)$$

By construction,

$$\bar{p}_{1,n}(z) = \mathbb{E}_X [p_{Z|X}(z|X)], \quad X \sim \mathcal{CN}(0, Q_n). \quad (46)$$

Define Willie's single-letter and n -letter information densities as

$$\iota_{w,n}(x; z) \triangleq \log \frac{p_{Z|X}(z|x)}{\bar{p}_{1,n}(z)}, \quad (47)$$

and

$$\iota_{w,n}^{(n)}(x^n; z^n) \triangleq \sum_{i=1}^n \iota_{w,n}(x_i; z_i). \quad (48)$$

The density of the reference distribution is therefore

$$\bar{p}_{1,n}^{(n)}(z^n) = \prod_{i=1}^n \bar{p}_{1,n}(z_i),$$

and the random codebook $\mathbf{C}_n$ induces

$$p_{1,\mathbf{C}_n}^{(n)}(z^n) = \frac{1}{M_n} \sum_{m=1}^{M_n} \prod_{i=1}^n p_{Z|X}(z_i | X_i(m)). \quad (49)$$

Hence,

$$\frac{p_{1,C_n}^{(n)}(z^n)}{\bar{p}_{1,n}^{(n)}(z^n)} = \frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left[\iota_{w,n}^{(n)}(X^n(m); z^n) \right]. \quad (50)$$

By symmetry of the i.i.d. random codebook, let $X^n(1)$ denote the transmitted codeword and generate Z^n according to $\prod_{i=1}^n p_{Z|X}(z_i|X_i(1))$. Then

$$\begin{aligned} \mathbb{E}_{C_n} D \left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) \\ = \mathbb{E} \log \left[\frac{1}{M_n} \sum_{m=1}^{M_n} \exp \left(\iota_{w,n}^{(n)}(X^n(m); Z^n) \right) \right]. \end{aligned} \quad (51)$$

For every $j \neq 1$, $X^n(j)$ is independent of Z^n , and

$$\begin{aligned} \mathbb{E}_{X^n(j)} \exp \left[\iota_{w,n}^{(n)}(X^n(j); z^n) \right] &= \prod_{i=1}^n \frac{\mathbb{E}_X [p_{Z|X}(z_i|X)]}{\bar{p}_{1,n}(z_i)} \\ &= 1. \end{aligned} \quad (52)$$

Conditioning on $(X^n(1), Z^n)$ and applying Jensen's inequality to $X^n(2), \dots, X^n(M_n)$ therefore gives

$$\begin{aligned} \mathbb{E}_{C_n} D \left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \mathbb{E} \log \left[\frac{e^{\iota_{w,n}^{(n)}(X^n; Z^n)} + M_n - 1}{M_n} \right] \\ &= \mathbb{E} \log \left[1 + \frac{e^{\iota_{w,n}^{(n)}(X^n; Z^n)} - 1}{M_n} \right] \\ &\leq \mathbb{E} \log \left[1 + \frac{e^{\iota_{w,n}^{(n)}(X^n; Z^n)}}{M_n} \right]. \end{aligned} \quad (53)$$

For any fixed $0 < \alpha < 1$, using $\log(1+t) \leq t^\alpha/\alpha$ for $t \geq 0$ yields

$$\begin{aligned} \mathbb{E}_{C_n} D \left(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)} \right) &\leq \frac{1}{\alpha M_n^\alpha} \mathbb{E} \exp \left[\alpha \iota_{w,n}^{(n)}(X^n; Z^n) \right] \\ &= \frac{1}{\alpha M_n^\alpha} \left(\mathbb{E} e^{\alpha \iota_{w,n}^{(n)}(X; Z)} \right)^n, \end{aligned} \quad (54)$$

where the last equality follows from the memoryless channel and the i.i.d. Gaussian input.

Gaussian Exponential Moment at Willie. Define

$$s_{w,n} \triangleq \frac{|g_{aw}|^2 Q_n}{N_w(P_d)}. \quad (55)$$

Under the joint distribution of (X, Z) , define

$$U \triangleq \frac{g_{aw}X}{|g_{aw}|\sqrt{Q_n}}, \quad W \triangleq \frac{Z - g_{aw}X}{\sqrt{N_w(P_d)}}.$$

Then

$$U, W \stackrel{\text{i.i.d.}}{\sim} \mathcal{CN}(0, 1), \quad \frac{Z}{\sqrt{N_w(P_d)}} = \sqrt{s_{w,n}}U + W.$$

Substitution into (42) gives

$$\iota_{w,n}(X; Z) = \log(1 + s_{w,n}) - |W|^2 + \frac{|\sqrt{s_{w,n}}U + W|^2}{1 + s_{w,n}}. \quad (56)$$

Let

$$\boldsymbol{\xi} \triangleq \begin{bmatrix} U \\ W \end{bmatrix} \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}_2).$$

Then

$$\iota_{w,n}(X; Z) = \log(1 + s_{w,n}) + \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}, \quad (57)$$

where

$$\mathbf{A}_{w,n} = \frac{1}{1 + s_{w,n}} \begin{bmatrix} s_{w,n} & \sqrt{s_{w,n}} \\ \sqrt{s_{w,n}} & -s_{w,n} \end{bmatrix}. \quad (58)$$

For $0 < \alpha < 1$, $\mathbf{I}_2 - \alpha \mathbf{A}_{w,n} \succ 0$, since

$$1 - \frac{\alpha s_{w,n}}{1 + s_{w,n}} > 0$$

and

$$\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n}) = 1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}} > 0. \quad (59)$$

Hence the proper-complex Gaussian integral gives

$$\begin{aligned} \mathbb{E} e^{\alpha \mathbf{z}_{w,n}(X;Z)} &= (1 + s_{w,n})^\alpha \mathbb{E} e^{\alpha \boldsymbol{\xi}^H \mathbf{A}_{w,n} \boldsymbol{\xi}} \\ &= \frac{(1 + s_{w,n})^\alpha}{\det(\mathbf{I}_2 - \alpha \mathbf{A}_{w,n})} \\ &= \frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}}. \end{aligned} \quad (60)$$

Substituting into (49),

$$\mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \leq \frac{1}{\alpha M_n^\alpha} \left[\frac{(1 + s_{w,n})^\alpha}{1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}} \right]^n. \quad (61)$$

Asymptotic Resolvability. Define Willie's Gaussian-input mutual information as

$$I_{w,n} \triangleq \log(1 + s_{w,n}). \quad (62)$$

The preceding resolvability bound can be written, for any $0 < \alpha < 1$, as

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \\ \leq \exp\left\{-\log \alpha - \alpha(\log M_n - nI_{w,n}) - n \log\left(1 - \frac{\alpha^2 s_{w,n}}{1 + s_{w,n}}\right)\right\}. \end{aligned} \quad (63)$$

From the selected power sequence,

$$s_{w,n} = \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}}, \quad (64)$$

so that

$$s_{w,n} = \Theta(n^{-1/2}), \quad ns_{w,n} = \Theta(\sqrt{n}). \quad (65)$$

Consider codebook sizes satisfying

$$\log M_n - nI_{w,n} \geq n^{3/8}. \quad (66)$$

Since the one-shot bound holds for every $0 < \alpha < 1$ at each n , choose

$$\alpha_n \triangleq \frac{n^{3/8}}{4ns_{w,n}}. \quad (67)$$

Then

$$\alpha_n = \Theta(n^{-1/8}) \longrightarrow 0, \quad (68)$$

and hence $0 < \alpha_n < 1$ for all sufficiently large n . Moreover,

$$\frac{\alpha_n^2 s_{w,n}}{1 + s_{w,n}} \longrightarrow 0.$$

Using $-\log(1 - u) \leq 2u$ for sufficiently small $u \geq 0$ gives

$$\begin{aligned} \mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}\right) \\ \leq \exp\left\{-\log \alpha_n - \alpha_n n^{3/8} + \frac{2n\alpha_n^2 s_{w,n}}{1 + s_{w,n}}\right\} \\ \leq \exp\left\{-\log \alpha_n - \alpha_n n^{3/8} + 2n\alpha_n^2 s_{w,n}\right\} \\ = \frac{4ns_{w,n}}{n^{3/8}} \exp\left(-\frac{n^{3/4}}{8ns_{w,n}}\right). \end{aligned} \quad (69)$$

Since $ns_{w,n} = \Theta(\sqrt{n})$,

$$\frac{4ns_{w,n}}{n^{3/8}} = O(n^{1/8}), \quad \frac{n^{3/4}}{8ns_{w,n}} = \Theta(n^{1/4}), \quad (70)$$

and therefore

$$\mathbb{E}_{\mathbf{C}_n} D\left(P_{1,\mathbf{C}_n}^{(n)} \parallel \overline{P}_{1,n}^{(n)}\right) \rightarrow 0. \quad (71)$$

Thus, a sufficient resolvability condition is

$$\log M_n \geq nI_{w,n} + n^{3/8}. \quad (72)$$

Compatibility and Codebook-Size Selection. The Gaussian-input mutual information at Willie satisfies

$$\begin{aligned} nI_{w,n} &= n \log \left(1 + \frac{|g_{aw}|^2 Q_n}{N_w(P_d)} \right) \\ &= \frac{|g_{aw}|^2}{N_w(P_d)} nQ_n + O(nQ_n^2) \\ &= (1 - \varepsilon)\sqrt{2\delta} \sqrt{n} + O(1). \end{aligned} \quad (73)$$

Combining this with the corresponding expansion of $nI_{b,n}$ gives

$$n(I_{b,n} - I_{w,n}) = (1 - \varepsilon)\sqrt{2\delta} \left[\frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} - 1 \right] \sqrt{n} + O(1). \quad (74)$$

Assume

$$\frac{|h_{ab}|^2}{N_b(P_d)} > \frac{|g_{aw}|^2}{N_w(P_d)}. \quad (75)$$

Then

$$n(I_{b,n} - I_{w,n}) = \Theta(\sqrt{n}), \quad (76)$$

with a strictly positive leading coefficient. Since $n^{3/8} = o(\sqrt{n})$, there exists n_0 such that, for all $n \geq n_0$,

$$n(I_{b,n} - I_{w,n}) \geq 2n^{3/8} + \log 2. \quad (77)$$

For such n , define

$$M_n \triangleq \left\lfloor \exp\left(nI_{b,n} - n^{3/8}\right) \right\rfloor. \quad (78)$$

Since $nI_{b,n} - n^{3/8} \rightarrow \infty$, for all sufficiently large n ,

$$nI_{b,n} - n^{3/8} - \log 2 \leq \log M_n \leq nI_{b,n} - n^{3/8}. \quad (79)$$

Consequently,

$$nI_{b,n} - \log M_n \geq n^{3/8}, \quad (80)$$

and

$$\begin{aligned} \log M_n - nI_{w,n} &\geq n(I_{b,n} - I_{w,n}) - n^{3/8} - \log 2 \\ &\geq n^{3/8}. \end{aligned} \quad (81)$$

Hence the selected M_n simultaneously satisfies the sufficient conditions for Bob reliability and Willie resolvability. Moreover,

$$\begin{aligned} \log M_n &= nI_{b,n} - n^{3/8} + O(1) \\ &= (1 - \varepsilon)\sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \end{aligned} \quad (82)$$

Per-Codeword Power Constraint. Define the event that all codewords in the random codebook satisfy the prescribed power constraint as

$$\mathcal{G}_n \triangleq \left\{ \frac{1}{n} \|X^n(m)\|^2 \leq P_a, \forall m \in \{1, \dots, M_n\} \right\}. \quad (83)$$

Its complement is therefore

$$\mathcal{G}_n^c = \left\{ \exists m \in \{1, \dots, M_n\} : \frac{1}{n} \|X^n(m)\|^2 > P_a \right\}. \quad (84)$$

Since $P_a > 0$ is fixed and $Q_n = \Theta(n^{-1/2})$,

$$\frac{P_a}{Q_n} \longrightarrow \infty. \quad (85)$$

For any fixed codeword index m , the Gaussian construction gives

$$\frac{|X_i(m)|^2}{Q_n} \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1), \quad i = 1, \dots, n,$$

and hence, for every $0 < t < 1$,

$$\mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) = \frac{1}{1-t}.$$

Applying the Chernoff bound,

$$\begin{aligned} \Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] &= \Pr\left[\sum_{i=1}^n \frac{|X_i(m)|^2}{Q_n} > \frac{nP_a}{Q_n}\right] \\ &\leq \exp\left(-\frac{tnP_a}{Q_n}\right) \prod_{i=1}^n \mathbb{E} \exp\left(t \frac{|X_i(m)|^2}{Q_n}\right) \\ &= \exp\left\{-n \left[\frac{tP_a}{Q_n} + \log(1-t)\right]\right\}. \end{aligned} \quad (86)$$

Choosing $t = 1/2$ yields

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left\{-n \left[\frac{P_a}{2Q_n} - \log 2\right]\right\}. \quad (87)$$

Since $P_a/Q_n \rightarrow \infty$, for all sufficiently large n ,

$$\frac{P_a}{4Q_n} \geq \log 2,$$

and therefore

$$\Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \leq \exp\left(-\frac{nP_a}{4Q_n}\right). \quad (88)$$

Using the definition of $\mathcal{G}_n^c$ and a union bound over all codewords,

$$\begin{aligned} \Pr(\mathcal{G}_n^c) &= \Pr\left[\bigcup_{m=1}^{M_n} \left\{\frac{1}{n}\|X^n(m)\|^2 > P_a\right\}\right] \\ &\leq \sum_{m=1}^{M_n} \Pr\left[\frac{1}{n}\|X^n(m)\|^2 > P_a\right] \\ &= M_n \Pr\left[\frac{1}{n}\|X^n(1)\|^2 > P_a\right] \\ &\leq \exp\left\{\log M_n - \frac{nP_a}{4Q_n}\right\}. \end{aligned} \quad (89)$$

For the previously selected codebook size,

$$\log M_n = O(\sqrt{n}), \quad (90)$$

whereas

$$Q_n = \Theta(n^{-1/2}) \implies \frac{nP_a}{Q_n} = \Theta(n^{3/2}). \quad (91)$$

Consequently,

$$\log M_n - \frac{nP_a}{4Q_n} = -\Theta(n^{3/2}), \quad (92)$$

and hence

$$\Pr(\mathcal{G}_n^c) \longrightarrow 0, \quad \Pr(\mathcal{G}_n) \longrightarrow 1. \quad (93)$$

Thus, with probability tending to one under the random-codebook ensemble, all codewords simultaneously satisfy

$$\frac{1}{n}\|X^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \quad (94)$$

Actual Coverttness. Let $p_0^{(n)}(z^n)$ denote the density of $P_0^{(n)}$. We first verify that the Gaussian reference distribution is exactly the ensemble-average output distribution induced by the random codebook. For every $z^n \in \mathbb{C}^n$,

$$\begin{aligned}\mathbb{E}_{C_n} [p_{1,C_n}^{(n)}(z^n)] &= \frac{1}{M_n} \sum_{m=1}^{M_n} \mathbb{E}_{X^n(m)} \left[\prod_{i=1}^n p_{Z|X}(z_i | X_i(m)) \right] \\ &= \prod_{i=1}^n \mathbb{E}_X [p_{Z|X}(z_i | X)] \\ &= \prod_{i=1}^n \bar{p}_{1,n}(z_i) \\ &= \bar{p}_{1,n}^{(n)}(z^n).\end{aligned}\tag{95}$$

Hence,

$$\mathbb{E}_{C_n} [P_{1,C_n}^{(n)}] = \bar{P}_{1,n}^{(n)}.\tag{96}$$

For every realization of the random codebook,

$$\begin{aligned}D(P_{1,C_n}^{(n)} \parallel P_0^{(n)}) &= D(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}) \\ &\quad + \int_{\mathbb{C}^n} p_{1,C_n}^{(n)}(z^n) \log \frac{\bar{p}_{1,n}^{(n)}(z^n)}{p_0^{(n)}(z^n)} dz^n.\end{aligned}\tag{97}$$

Taking expectation over C_n and using the preceding ensemble-average identity gives

$$\begin{aligned}\mathbb{E}_{C_n} D(P_{1,C_n}^{(n)} \parallel P_0^{(n)}) \\ = \mathbb{E}_{C_n} D(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}) + D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}).\end{aligned}\tag{98}$$

Since

$$s_{w,n} = (1 - \varepsilon) \sqrt{\frac{2\delta}{n}},$$

the reference divergence is

$$D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}) = n [s_{w,n} - \log(1 + s_{w,n})].\tag{99}$$

For every $x \geq 0$,

$$x - \log(1 + x) \leq \frac{x^2}{2}.\tag{100}$$

Indeed,

$$\frac{d}{dx} \left[\frac{x^2}{2} - x + \log(1 + x) \right] = \frac{x^2}{1 + x} \geq 0,$$

and the expression in brackets equals zero at $x = 0$. Therefore,

$$\begin{aligned}D(\bar{P}_{1,n}^{(n)} \parallel P_0^{(n)}) &\leq \frac{ns_{w,n}^2}{2} \\ &= (1 - \varepsilon)^2 \delta.\end{aligned}\tag{101}$$

Moreover, the preceding resolvability result gives

$$\mathbb{E}_{C_n} D(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}) \longrightarrow 0.\tag{102}$$

Since

$$\delta - (1 - \varepsilon)^2 \delta = \delta(2\varepsilon - \varepsilon^2) > 0,\tag{103}$$

there exists n_0 such that, for every $n \geq n_0$,

$$\mathbb{E}_{C_n} D(P_{1,C_n}^{(n)} \parallel \bar{P}_{1,n}^{(n)}) \leq \frac{\delta(2\varepsilon - \varepsilon^2)}{2}.\tag{104}$$

Consequently,

$$\begin{aligned}\mathbb{E}_{C_n} D(P_{1,C_n}^{(n)} \parallel P_0^{(n)}) &\leq (1 - \varepsilon)^2 \delta + \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &= \delta - \frac{\delta(2\varepsilon - \varepsilon^2)}{2} \\ &< \delta.\end{aligned}\tag{105}$$

Finally, since KL divergence is nonnegative, Markov's inequality gives, for all sufficiently large n ,

$$\begin{aligned} & \Pr \left[D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) > \delta \right] \\ & \leq \frac{\mathbb{E}_{\mathcal{C}_n} D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right)}{\delta} \\ & \leq 1 - \frac{2\varepsilon - \varepsilon^2}{2}. \end{aligned} \quad (106)$$

Hence,

$$\Pr \left[D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) \leq \delta \right] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (107)$$

Thus, for all sufficiently large n , the random-codebook ensemble contains codebook realizations satisfying the actual covertness constraint with probability bounded away from zero.

Deterministic Fixed-Codebook Extraction. For a realization $\mathcal{C}_n$ of the random codebook, let $P_e^{(n)}(\mathcal{C}_n)$ denote the average decoding error probability of its associated ML decoder. The preceding reliability analysis gives

$$\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right] \rightarrow 0. \quad (108)$$

Since $P_e^{(n)}(\mathcal{C}_n) \geq 0$, Markov's inequality yields

$$\begin{aligned} & \Pr \left[P_e^{(n)}(\mathcal{C}_n) > \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]} \right] \\ & \leq \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]} \rightarrow 0. \end{aligned} \quad (109)$$

The power-constraint analysis has established

$$\Pr(\mathcal{G}_n^c) \rightarrow 0, \quad (110)$$

where $\mathcal{G}_n$ is the event that every codeword satisfies the per-codeword power constraint. Moreover, the actual-covertness analysis gives, for all sufficiently large n ,

$$\Pr \left[D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) \leq \delta \right] \geq \frac{2\varepsilon - \varepsilon^2}{2} > 0. \quad (111)$$

Using the union bound on their complements,

$$\begin{aligned} & \Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathcal{C}_n) \leq \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]}, \right. \\ & \quad \left. D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) \leq \delta \right] \\ & \geq \Pr \left[D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) \leq \delta \right] \\ & \quad - \Pr(\mathcal{G}_n^c) \\ & \quad - \Pr \left[P_e^{(n)}(\mathcal{C}_n) > \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]} \right] \\ & \geq \frac{2\varepsilon - \varepsilon^2}{2} - \Pr(\mathcal{G}_n^c) - \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]}. \end{aligned} \quad (112)$$

The last two terms vanish as $n \rightarrow \infty$, whereas $(2\varepsilon - \varepsilon^2)/2 > 0$. Hence, for all sufficiently large n ,

$$\Pr \left[\mathcal{G}_n, P_e^{(n)}(\mathcal{C}_n) \leq \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]}, D \left(P_{1, \mathcal{C}_n}^{(n)} \parallel P_0^{(n)} \right) \leq \delta \right] > 0. \quad (113)$$

Therefore, for every sufficiently large n , there exists a deterministic realization

$$\mathcal{C}_n^* \quad (114)$$

of the random codebook such that

$$P_e^{(n)}(\mathcal{C}_n^*) \leq \sqrt{\mathbb{E}_{\mathcal{C}_n} \left[P_e^{(n)}(\mathcal{C}_n) \right]} \rightarrow 0, \quad (115)$$

$$D\left(P_{1,\mathcal{C}_n^*}^{(n)} \parallel P_0^{(n)}\right) \leq \delta, \quad (116)$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \quad (117)$$

Thus $\{\mathcal{C}_n^*\}$ forms a sequence of deterministic fixed codebooks satisfying reliability, actual covertness, and the per-codeword power constraint simultaneously.

Since every realization of $\mathcal{C}_n$ contains exactly M_n codewords, the extraction does not alter the previously selected codebook size. Hence

$$\log M_n = (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \quad (118)$$

Achievable Square-Root Coefficient. For every fixed $0 < \varepsilon < 1$, the preceding construction yields a sequence of deterministic fixed codebooks $\{\mathcal{C}_n^*\}$ satisfying, for all sufficiently large n ,

$$P_e^{(n)}(\mathcal{C}_n^*) \rightarrow 0, \quad (119)$$

$$D\left(P_{1,\mathcal{C}_n^*}^{(n)} \parallel P_0^{(n)}\right) \leq \delta, \quad (120)$$

and

$$\frac{1}{n} \|x^n(m)\|^2 \leq P_a, \quad m = 1, \dots, M_n. \quad (121)$$

Moreover, deterministic-codebook extraction does not alter the previously selected codebook size, and hence

$$\begin{aligned} \log M_n &= nI_{b,n} - n^{3/8} + O(1) \\ &= (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)} \sqrt{n} + o(\sqrt{n}). \end{aligned} \quad (122)$$

Therefore,

$$\lim_{n \rightarrow \infty} \frac{\log M_n}{\sqrt{n}} = (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \quad (123)$$

Since $\mathcal{C}_n^*$ is an admissible deterministic codebook, the optimal codebook size satisfies

$$M_n^*(\delta) \geq M_n \quad (124)$$

for all sufficiently large n . Consequently,

$$\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq (1 - \varepsilon) \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}. \quad (125)$$

The above inequality holds for every fixed $0 < \varepsilon < 1$. Taking the supremum over ε and using

$$\sup_{0 < \varepsilon < 1} (1 - \varepsilon) = 1$$

gives

$$\boxed{\liminf_{n \rightarrow \infty} \frac{\log M_n^*(\delta)}{\sqrt{n}} \geq \sqrt{2\delta} \frac{|h_{ab}|^2 N_w(P_d)}{|g_{aw}|^2 N_b(P_d)}}. \quad (126)$$

□